APOLLO HOSPITALS

EXPLORATORY DATA ANALYSIS | PYTHON

Appointment No-Show and Patient Engagement Analysis

Booking behaviour, no-show patterns, patient engagement, financial performance and doctor service quality (2022–2024)

75,000	320	15	6
Appointments analysed	Doctors in dataset	Cities across India	Analysis sections

SUBMITTED BY
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Apollo Hospitals | Appointment Analytics

BACKGROUND & OBJECTIVE

Background

15
Cities across India

3
Consultation modes

5
Booking channels

- **Consultation modes:** In-Clinic, Video Consult, Home Visit
- **Booking channels:** Apollo App, Website, Call Centre, Walk-In, Partner App
- **Dataset:** 75,000 appointments + 320 doctors

The Problem

11,584
No-show appointments (15.4% of all bookings)

₹1.55 Cr
Potential revenue lost to no-shows

- **Doctor utilisation:** empty slots waste valuable doctor time
- **Revenue:** every missed visit is lost consultation income
- **Patient outcomes:** missed visits delay treatment and follow-up

Objective

Perform Exploratory Data Analysis in Python to turn Apollo's appointment data into actionable insights on booking behaviour, no-show patterns, patient engagement, financial performance and doctor service quality.

Goal: find where no-shows happen and how Apollo can reduce them and protect revenue.

What We Analyse

1. **Business Overview**
   Appointment trends, outcomes, channels
2. **No-Show Analysis**
   Specialty, city, lead time, channel
3. **Reminder & Engagement**
   Reminders, history, membership
4. **Patient & Demographic Segmentation**
   Age, gender, conditions, visit reasons
5. **Financial Performance**
   Revenue loss, cities, payment, insurance
6. **Doctor Utilisation & Service Quality**
   Utilisation, waiting time, satisfaction

DATASET & METHOD

Dataset

FACT TABLE

apollo_appointments_fact.csv

75,000 rows × 52 columns

One row = one appointment

Jan 2022 to Dec 2024

DOCTORS DIM

apollo_doctors_dim.csv

320 rows × 15 columns

One row = one doctor

Experience, specialty, rating

JOINED ON doctor_id

2022–2024	32,705	15 | 14
Appointment period	Unique patients	Cities | Specialties

Appointment Outcomes (75,000)



Completed	No-Show	Cancelled	Scheduled
55,275	11,584	5,793	2,348

72,652 eligible for no-show rate (Completed + No-Show + Cancelled)

Method & Rules Followed

1. **No-show rate**
   No-Show ÷ (Completed + No-Show + Cancelled). Scheduled is excluded, giving 72,652 eligible appointments and a 15.9% overall rate.
2. **Quality metrics**
   Wait time, consultation duration, satisfaction and utilisation are null for the 19,725 non-Completed rows, so they are analysed on Completed only.
3. **Revenue**
   Revenue columns are zero for non-Completed rows, so all revenue analysis is scoped to Completed only.
4. **“None” values**
   None in chronic condition, membership type, insurance provider and cancellation reason means not applicable, not missing.

Python Workflow

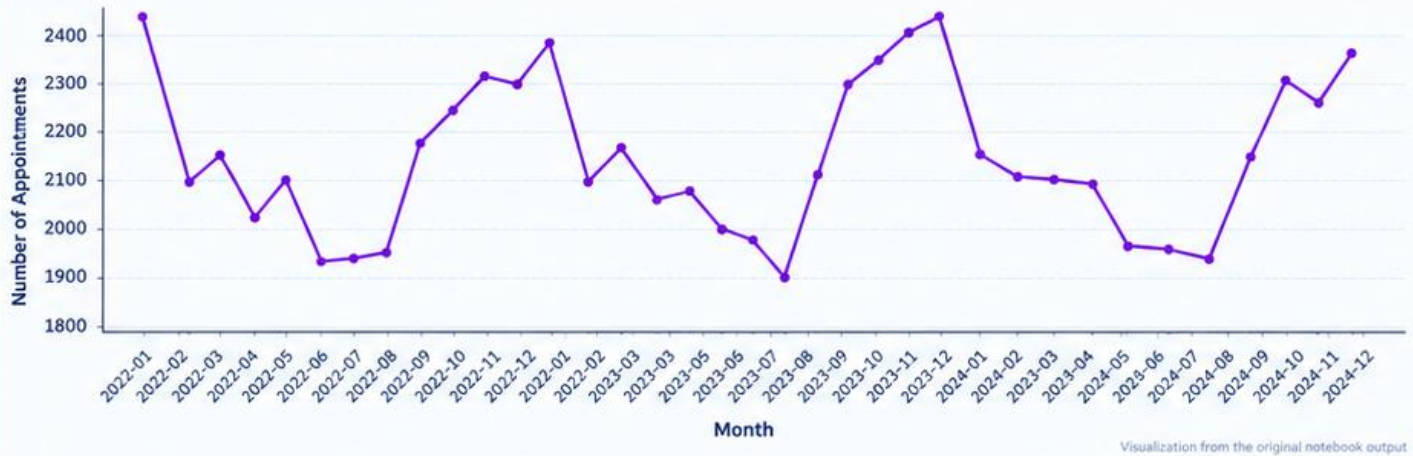
1. Load CSVs
2. Convert dates
3. Filter by rules
4. Group & pivot
5. Merge on doctor_id
6. Visualise

pandas · NumPy · Matplotlib · Seaborn

BUSINESS OVERVIEW

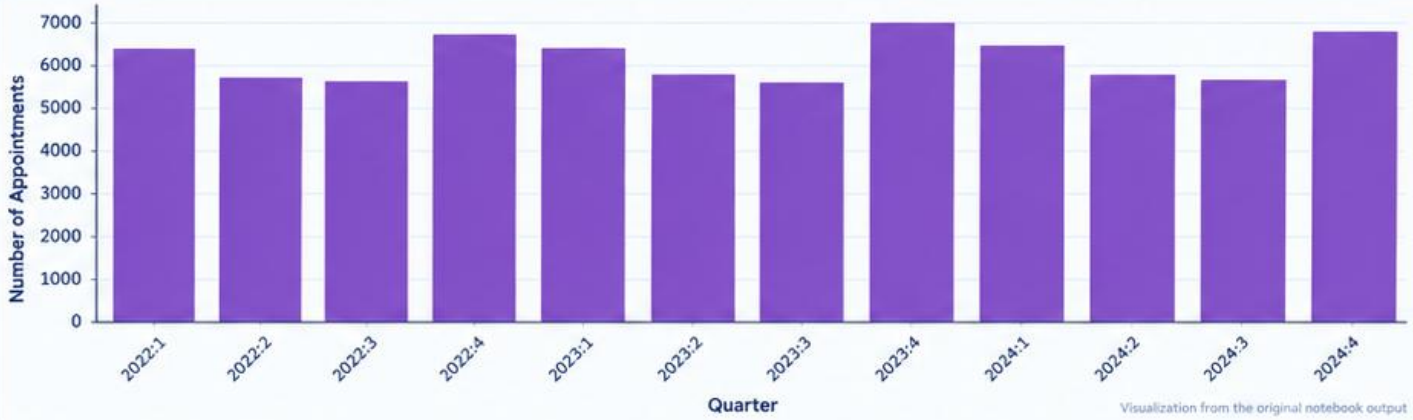
1. How is appointment volume trending across months and quarters from 2022 to 2024?

Monthly Appointment Volume (2022–2024)



Visualization from the original notebook output

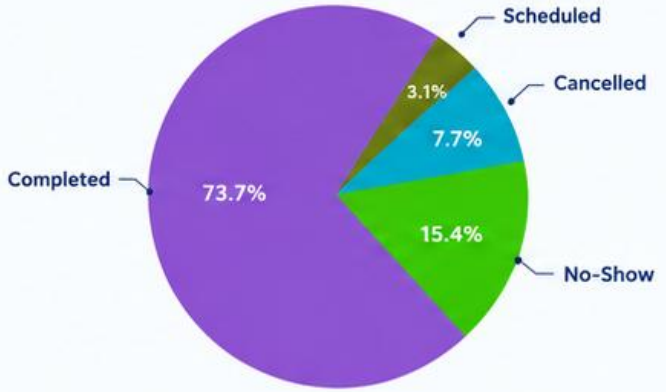
Quarterly Appointment Volume (2022–2024)



Visualization from the original notebook output

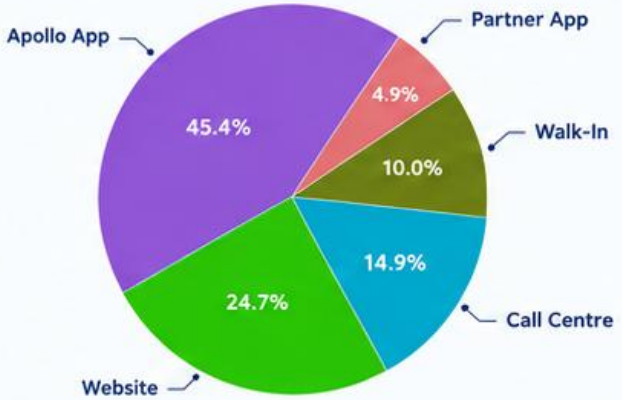
2. What is the split of outcomes across completed, no-show, cancelled, and scheduled appointments?

Distribution of Appointment Outcomes



3. Which booking channels drive the highest volume and how does the channel mix look?

Booking Channel Outcomes



NO-SHOW ANALYSIS

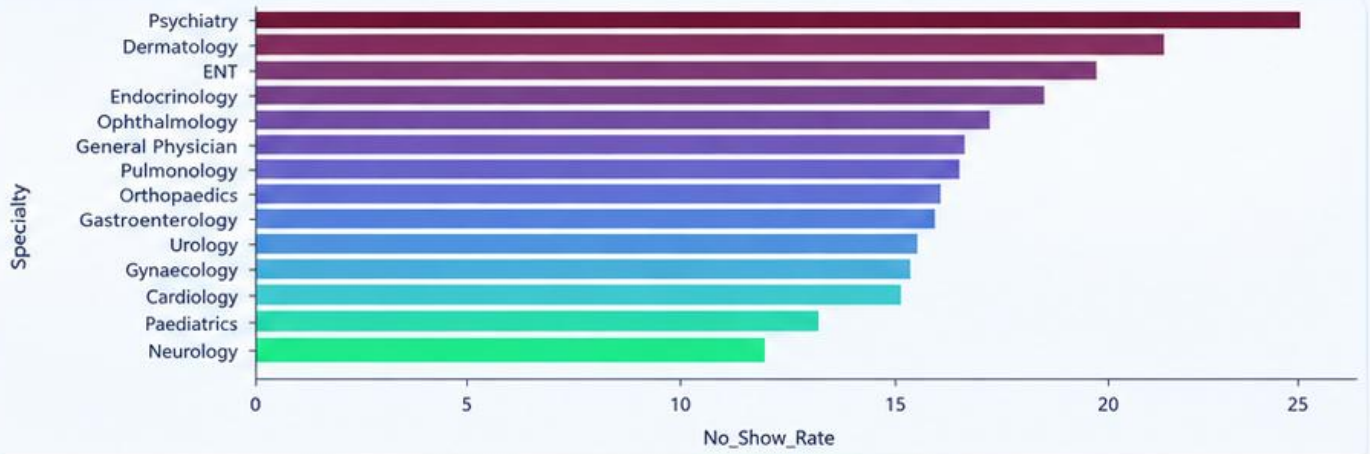
1. Which specialties and cities have the highest and lowest no-show rates?

specialty	Total_Eligible	Total_No_Shows	No_Show_Rate
Psychiatry	2900	711	24.517241
Dermatology	7566	1608	21.252974
ENT	5211	988	18.959893
Endocrinology	2024	355	17.539526
Ophthalmology	3103	503	16.210119
General Physician	19654	3049	15.513381
Pulmonology	2549	394	15.457042
Orthopaedics	5419	810	14.947407
Gastroenterology	2527	377	14.918876
Urology	2222	320	14.401440
Gynaecology	3991	569	14.257078
Cardiology	4484	629	14.027654
Paediatrics	6185	756	12.223120
Neurology	4817	515	10.691302

No-Show Rate by City



No-Show Rate by Specialty



NO-SHOW ANALYSIS

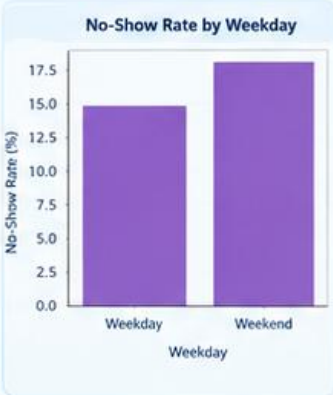
2. How does booking lead time affect the probability of a no-show?

Lead_Time_Group	Total_Eligible	Total_No_Shows	No_Show_Probability
0–1 days	10453	1382	13.221085
2–7 days	30919	4892	15.821986
8–14 days	8758	1648	18.817082
15–30 days	2342	469	20.025619
More than 30 days	0	0	NaN



3. Do evening slots, weekends, and longer-lead bookings see higher dropout?



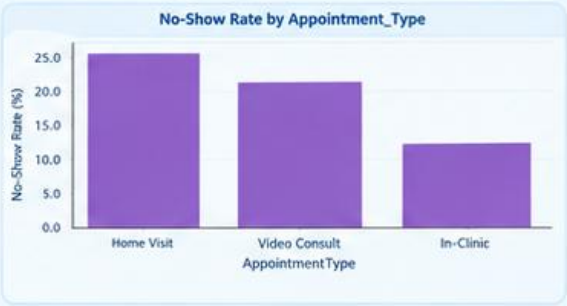




Visualization from the original notebook output

4. Which appointment types and booking channels carry the most no-show risk?

booking_channel	Total_channel	Total_no_shows	no_showRate
Walk-In	7281	1520	20.876253
Partner App	3594	651	18.113523
Website	17964	3090	17.201069
Call Centre	10855	1650	15.200368
Apollo App	32958	4673	14.178652



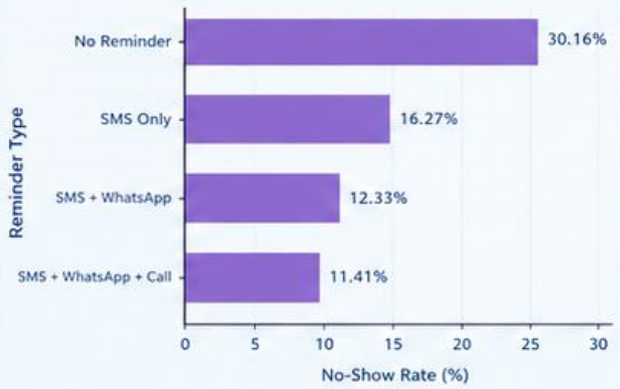


Visualization from the original notebook output

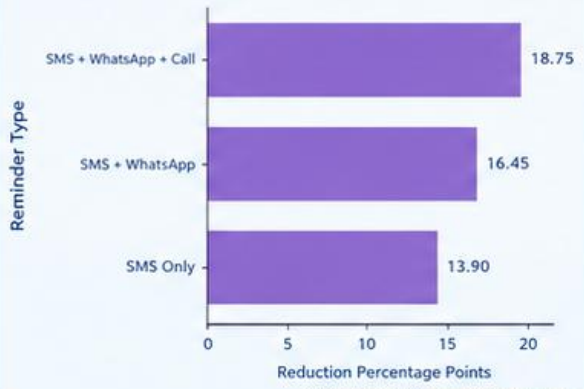
REMINDER AND ENGAGEMENT EFFECTIVENESS

1. How much does reminder type reduce no-show rates compared to no-reminder patients?

No-Show Rate by Reminder Type



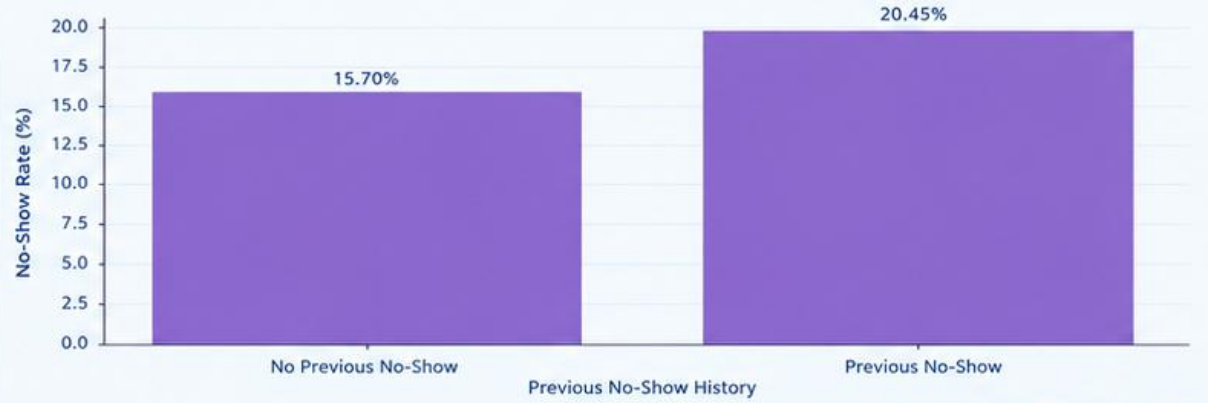
No-Show Rate Reduction vs No Reminder



Visualization from the original notebook output

2. Does prior no-show history predict future no-show behaviour?

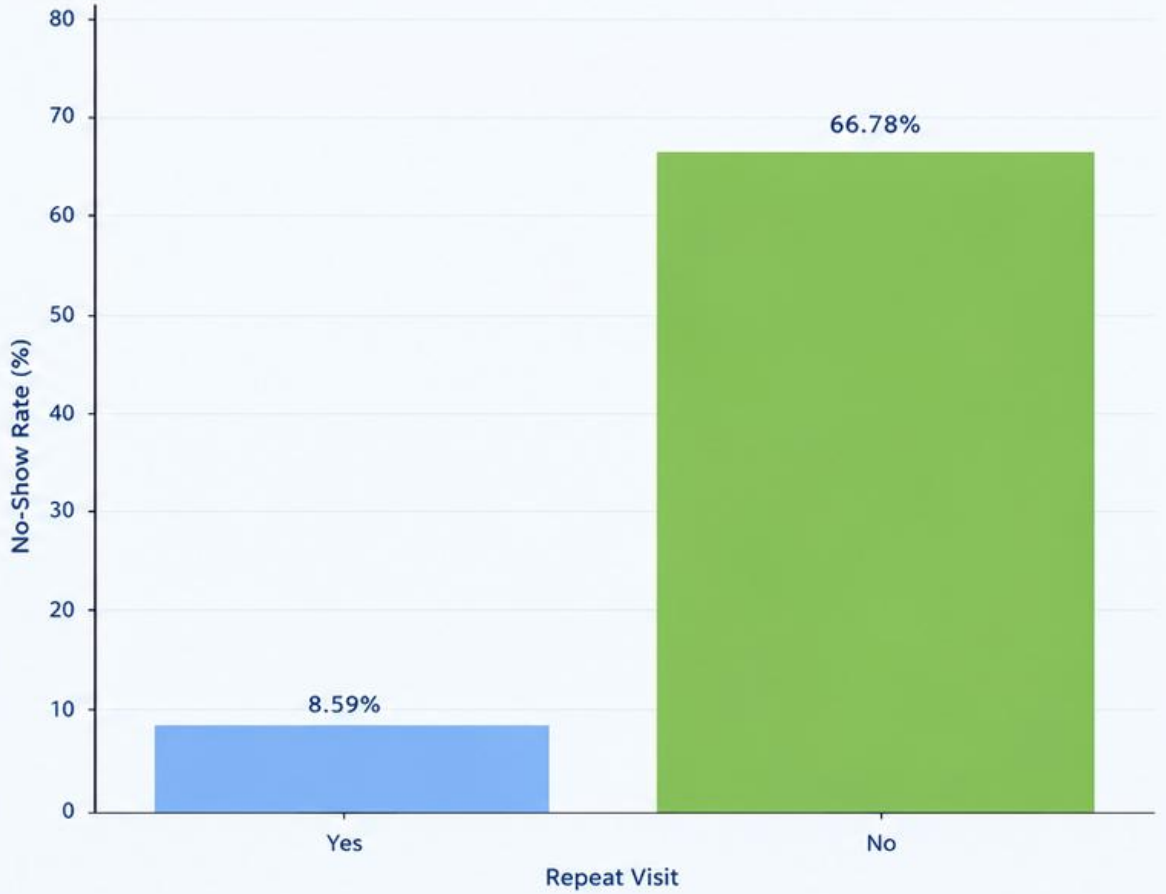
Prior No-Show History vs Future No-Show Rate



Visualization from the original notebook output

3. Do Apollo members and repeat patients show better attendance than non-members and first-time patients?

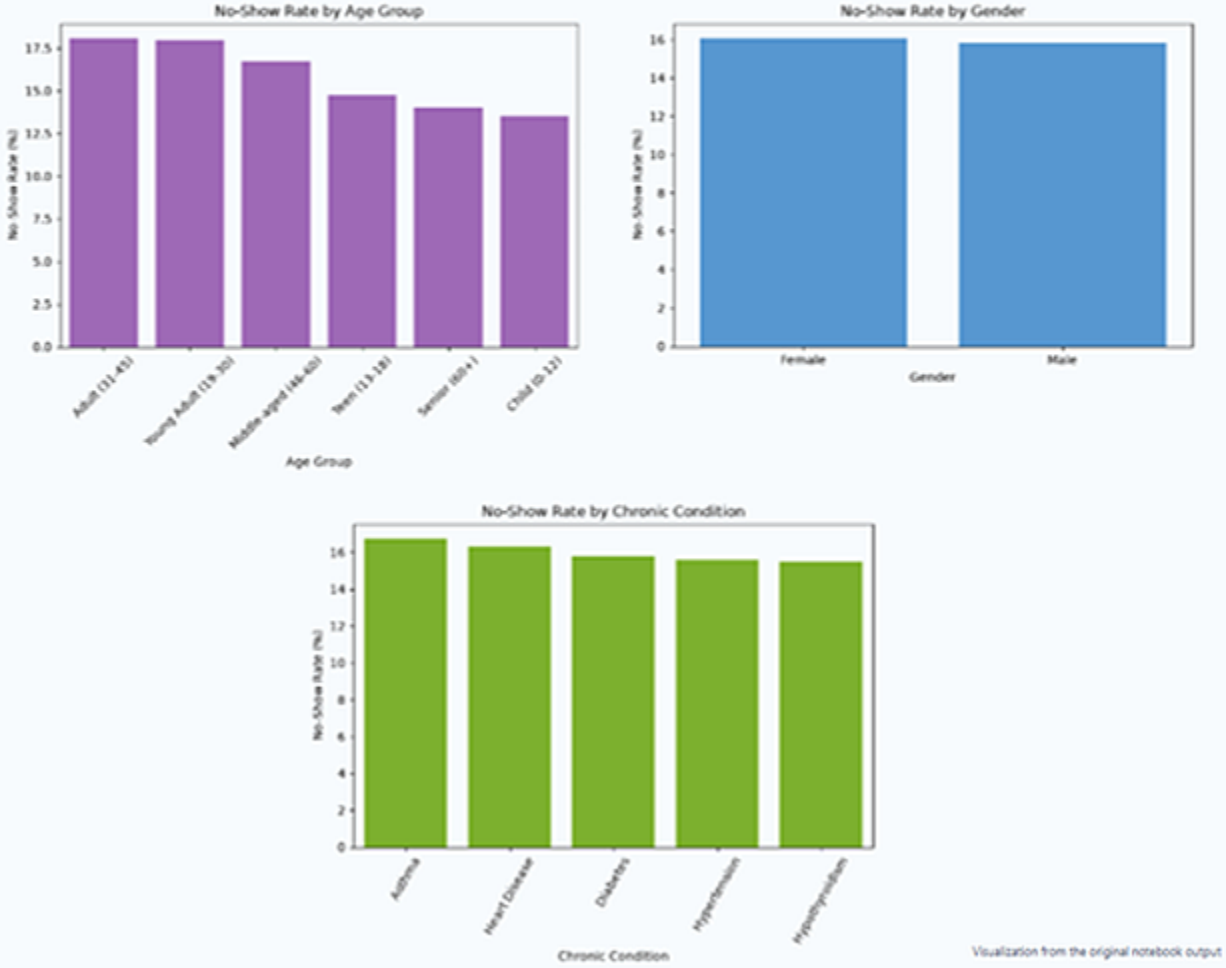
No-Show Rate by Repeat Visit Status



Visualization from the original notebook output

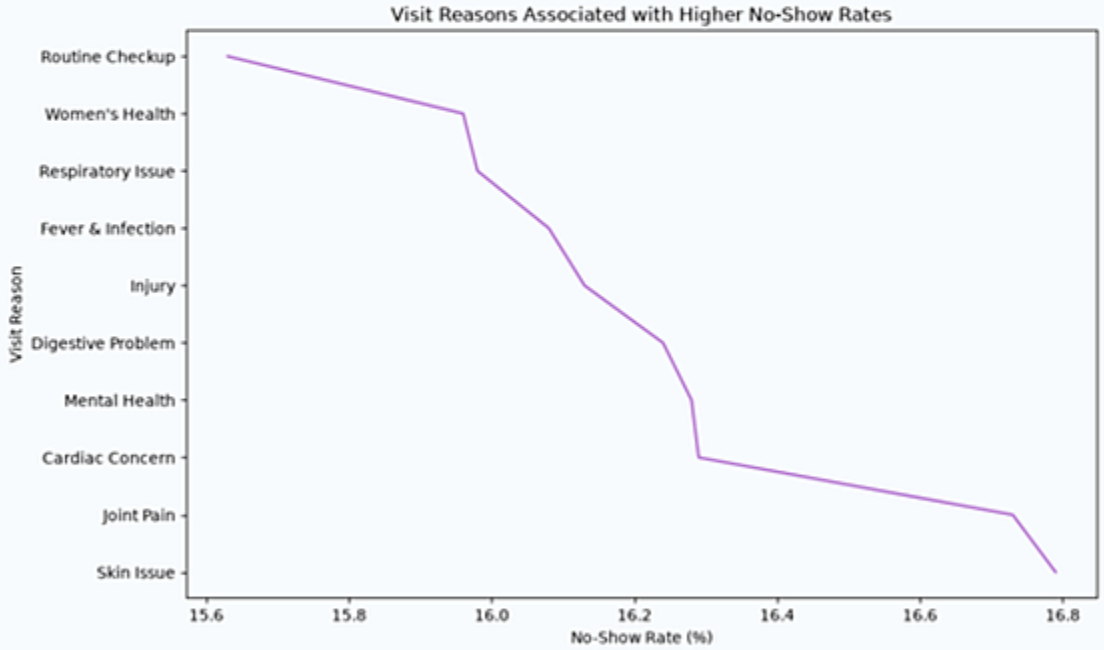
PATIENT & DEMOGRAPHIC SEGMENTATION

1. How do age group, gender, and chronic condition status influence no-show likelihood?



Visualization from the original notebook output

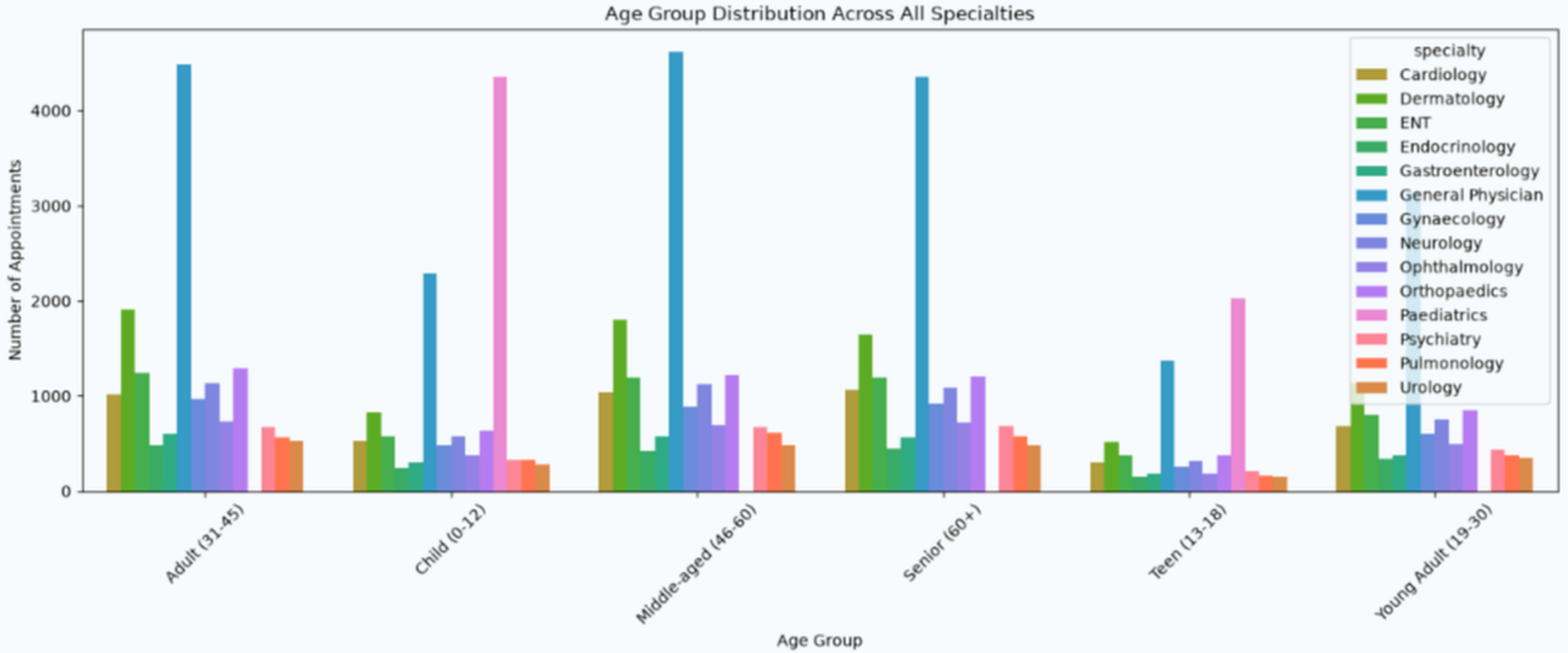
2. Which visit reasons are most common and are some associated with higher dropout?



Visualization from the original notebook output

PATIENT & DEMOGRAPHIC SEGMENTATION

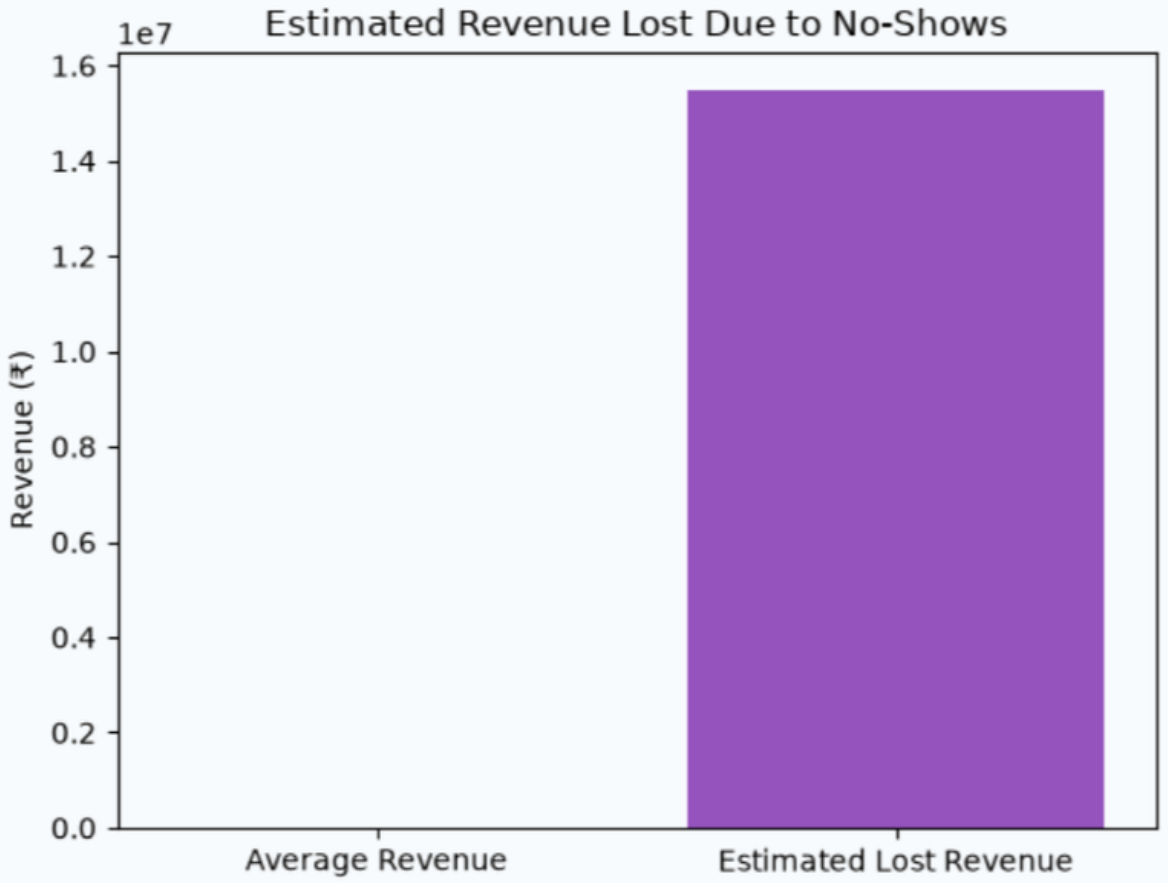
3. What does the patient age distribution look like within specialties like Paediatrics, Gynaecology, and Psychiatry?



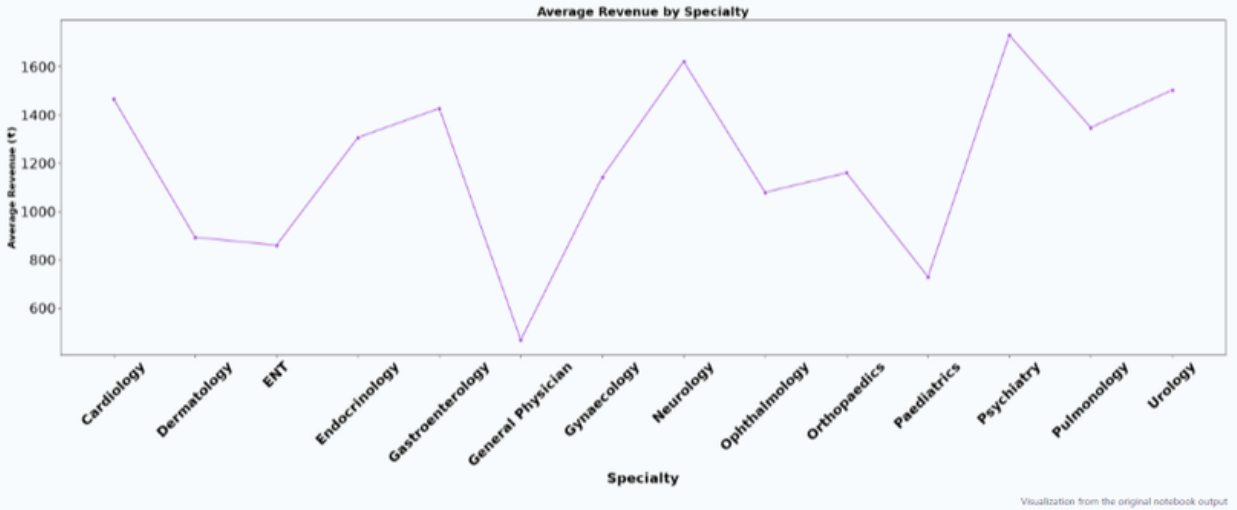
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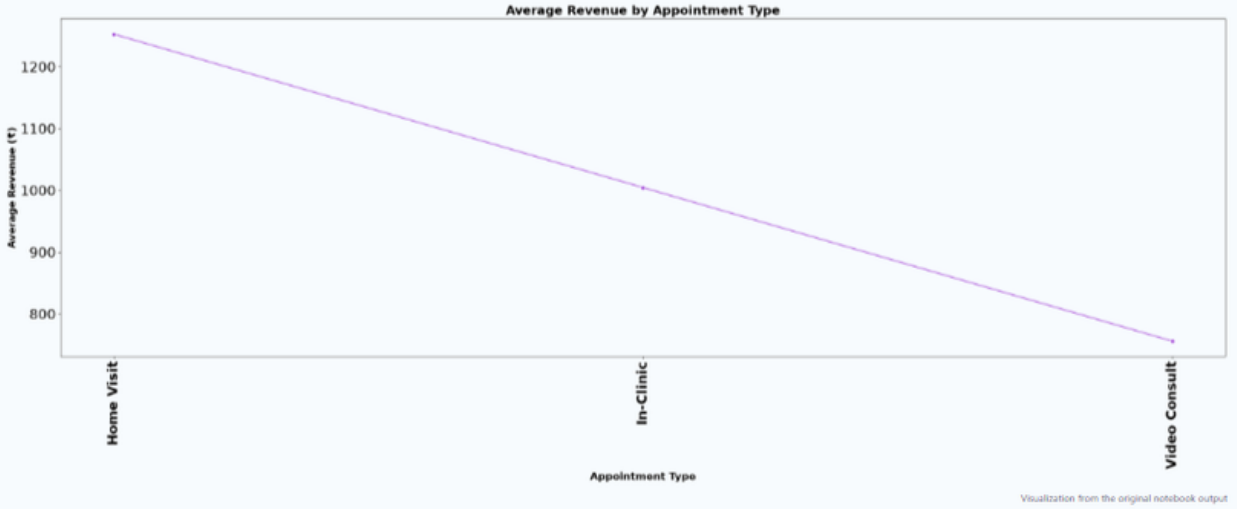
FINANCIAL PERFORMANCE

1. How much revenue is being lost to no-shows?



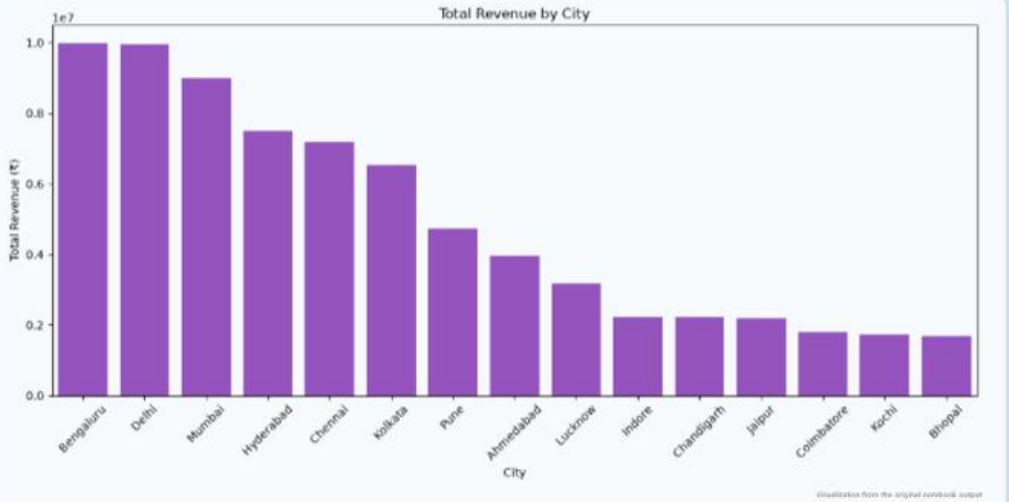
2. How does average revenue vary across specialties and appointment types?

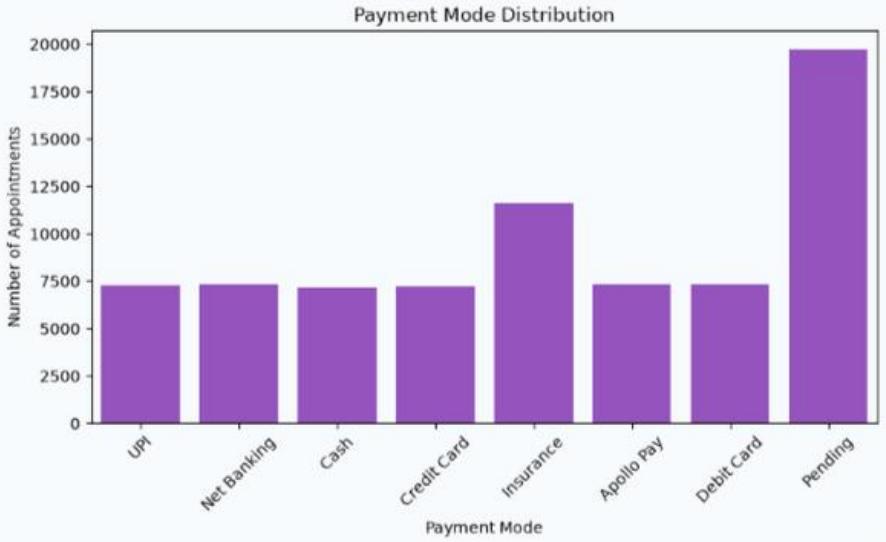




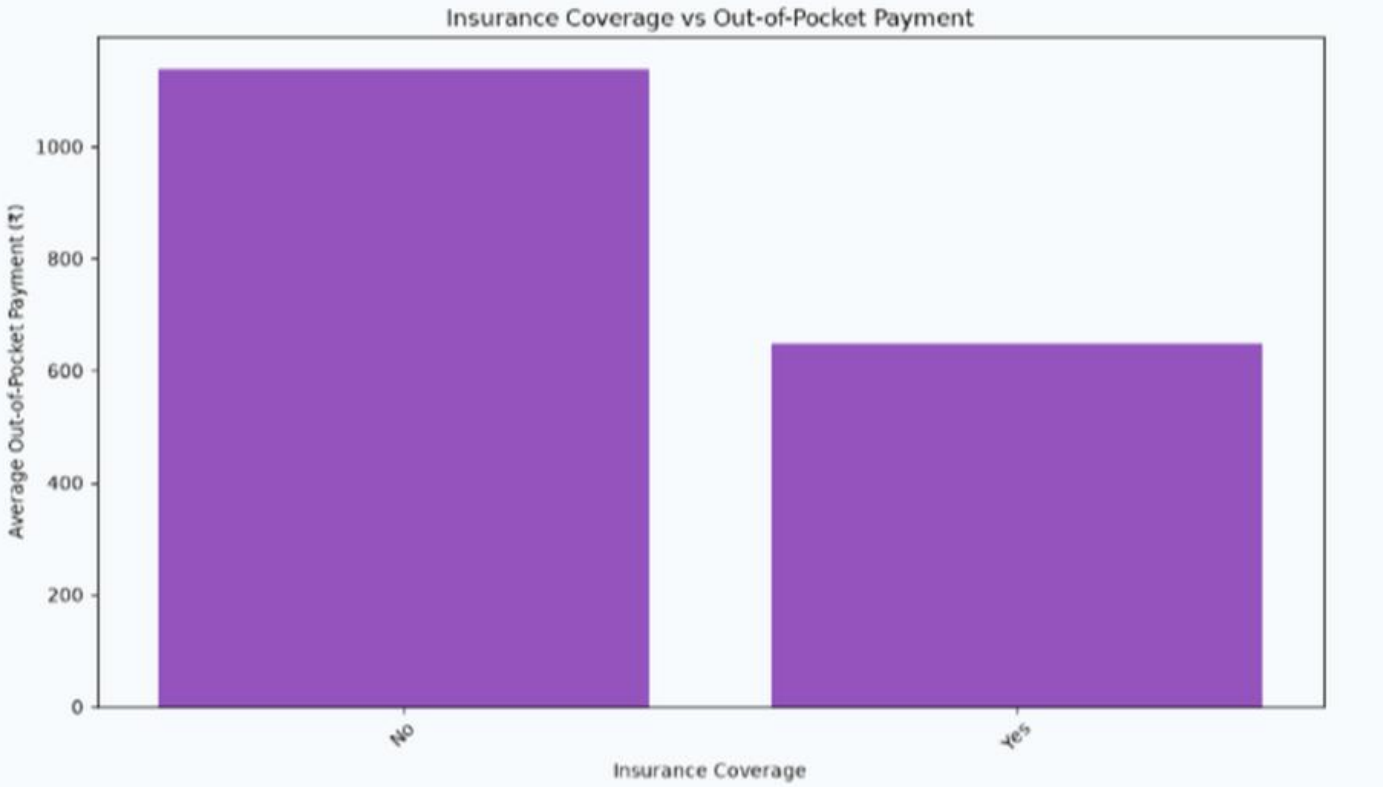
FINANCIAL PERFORMANCE

3. Which cities generate the most revenue and what is the payment mode mix?



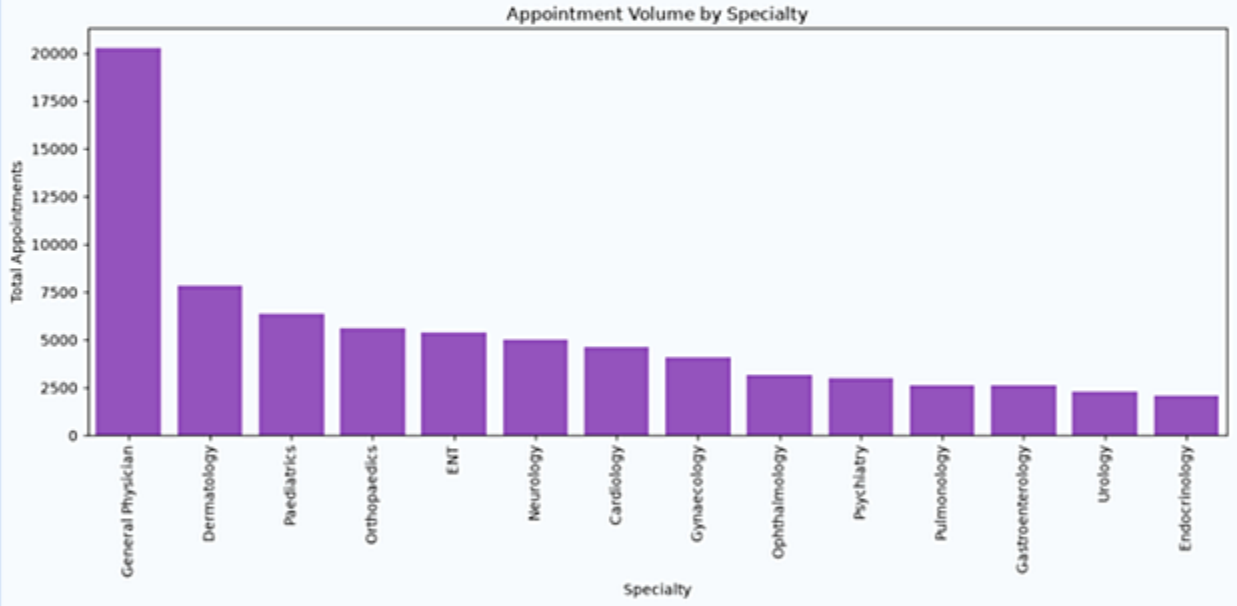


4. How does insurance coverage affect what patients pay out of pocket?



DOCTOR UTILIZATION & SERVICE QUALITY

1. Which specialties have the highest and lowest doctor utilisation rates?

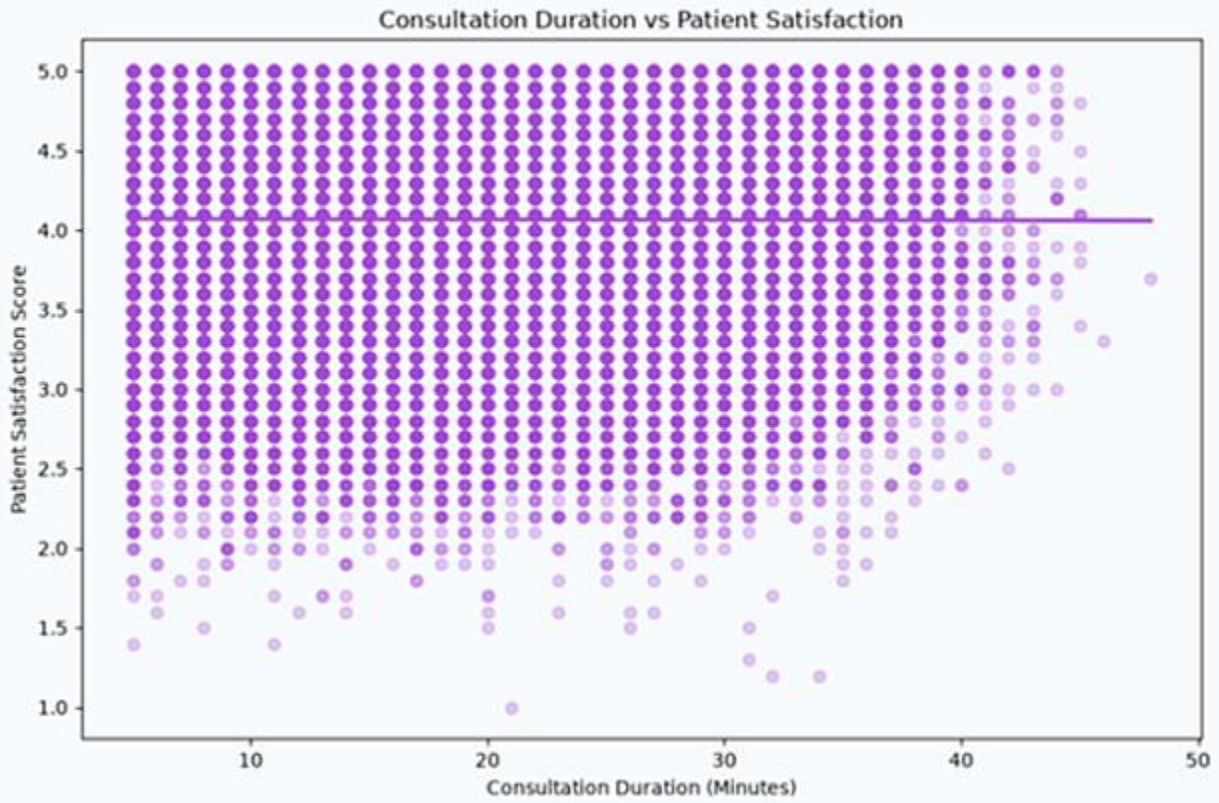


2. How long are patients waiting and does it vary by specialty or time of day?

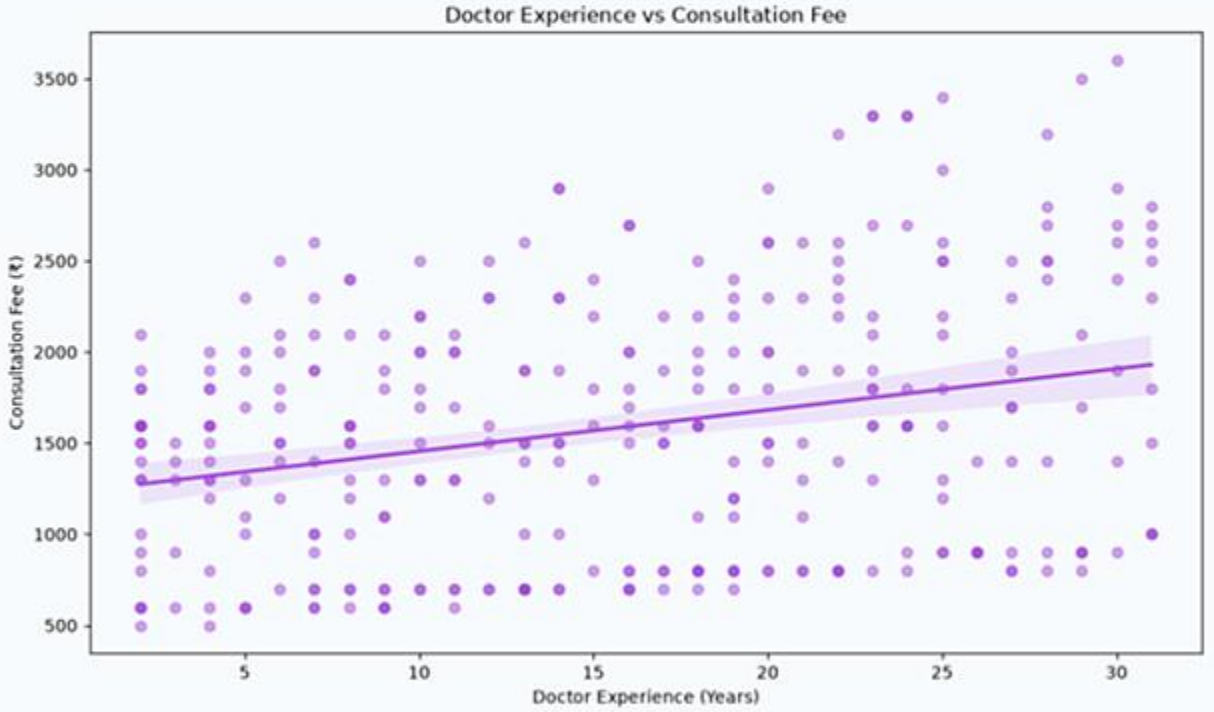


DOCTOR UTILIZATION & SERVICE QUALITY

3. Is there a relationship between consultation duration and patient satisfaction?



4. How does doctor experience relate to the fee charged (join with doctors dim)?



KEY INSIGHTS

1 Business Overview

- **73.7% completed** of 75,000 appointments; 15.4% (11,584) end as no-shows and 7.7% are cancelled.
- **Q4 is the peak quarter every year** (6,780 in 2022, 6,998 in 2023, 6,864 in 2024); Q2 and Q3 are the lows.
- **Apollo App drives 45.4% of bookings,** followed by Website (24.7%) and Call Centre (14.9%).

2 No-Show Analysis

- **Psychiatry (24.5%) and Dermatology (21.3%)** have the highest no-show rates; Neurology is lowest at 10.7%.
- **Longer lead time, higher risk:** 13.2% at 0–1 days rising to 20.0% at 15–30 days.
- **Riskiest bookings:** Home Visit (26.3%), Evening slots (18.4%), weekends (18.7% vs 14.8%) and Walk-In (20.9%).

3 Reminder & Engagement

- **Reminders work:** no-reminder patients no-show 30.2% vs 11.4% with SMS + WhatsApp + Call (18.75 points lower).
- **Past behaviour predicts the future:** 20.45% no-show with a previous no-show vs 15.70% without.
- **Repeat patients attend far more:** 8.6% vs 66.8% for first-timers; members 12.7% vs non-members 16.9%.

4 Patient & Demographic Segmentation

- **Ages 19–45 no-show most** (Adult 31–45: 18.1%, Young Adult: 17.9%); Children are lowest at 13.5%.
- **Gender and chronic condition matter little:** Female 16.0% vs Male 15.8%; conditions range only 15.5% to 16.7%.
- **Skin Issue (16.8%) and Joint Pain (16.7%)** visits carry the highest no-show rates; Follow-up is the most common visit reason (8,807).

5 Financial Performance

- **₹1.55 Cr potential revenue lost:** 11,584 no-shows × ₹1,337 average revenue per completed appointment.
- **Psychiatry earns most (₹1,730), General Physician least (₹468);** Home Visit (₹1,253) beats In-Clinic (₹1,004) and Video (₹756).
- **Bengaluru and Delhi lead city revenue (about ₹1.0 Cr each);** insured patients pay ₹649 out of pocket vs ₹1,138 without (43% less).

6 Doctor Utilisation & Service Quality

- **General Physician is busiest (20,272 appointments),** Endocrinology has the fewest (2,087).
- **Waiting time averages 11.5 minutes** and barely varies by specialty or time of day (10.6 to 12.8 minutes).
- **Longer consultations do not raise satisfaction** (correlation 0.0, average score 4.07), but fees rise with experience (about ₹23 per year).

RECOMMENDATIONS & NEXT STEPS

Recommendations (backed by the data)

1 Reminders for every patient

Make SMS + WhatsApp + Call the default reminder, so no appointment goes without one.

No reminder 30.2% → SMS + WhatsApp + Call 11.4%

2 Confirm long-lead bookings

Add a confirmation step for bookings made 8 or more days ahead.

13.2% at 0–1 days vs 20.0% at 15–30 days

3 Focus on high-risk segments

Prioritise reminders and confirmations for Psychiatry, Dermatology, Home Visit and Video Consult.

Psychiatry 24.5% · Home Visit 26.3% · In-Clinic 12.2%

4 Flag repeat no-show patients

Give extra follow-up to patients with a previous no-show and to first-time patients.

Prior no-show 20.45% vs 15.70% · First-time 66.8% vs repeat 8.6%

5 Fix risky channels and slots

Shift bookings to the Apollo App; add reminders for Walk-In, Partner App, evening and weekend slots.

Walk-In 20.9% · Partner App 18.1% vs Apollo App 14.2%

6 Grow Apollo membership

Promote Silver and Gold membership, since members attend more.

Members 12.7% vs non-members 16.9%

Next Steps

- Track no-show rate every month overall (15.9% now) and by specialty, lead time and channel.
- A/B test the multi-channel reminder on high-risk segments to confirm its effect.
- Build a no-show prediction model using lead time, reminder type, prior no-shows and visit history.
- Start with Psychiatry: highest no-show rate (24.5%) and highest average revenue (₹1,730).

Data Limitations

- Scheduled appointments (2,348) are excluded from the no-show rate.
- Wait time, satisfaction and revenue cover Completed appointments only.
- Revenue loss is an estimate: no-shows × average revenue per completed appointment.
- Results show association, not proven cause; no bookings exist in the 30+ days lead group.

APOLLO HOSPITALS

APPOINTMENT NO-SHOW AND PATIENT ENGAGEMENT ANALYSIS

Thank You

Questions & Discussion

15.4%
of all appointments end as no-shows

₹1.55 Cr
potential revenue lost to no-shows

30.2% → 11.4%
no-show rate with full reminders

PRESENTED BY
CHETAN AGRAWAL

LIVE DASHBOARD
chetan26agrawal.github.io/apollo-appointment-no-show-analysis